Example 10.2

Find the slope of tangent line to the graph of $f(x) = -5x^2 + 7x$ at $(5, f(5))$.

Solution

Step (i) $f(5) = -5(5)^2 + 7 \times 5 = -125 + 35 = -90$.

For any $\Delta x \neq 0$,

$$f(5 + \Delta x) = -5(5 + \Delta x)^2 + 7(5 + \Delta x) = -90 - 43\Delta x - 5(\Delta x)^2.$$

Step (ii) $$\Delta y = f(5 + \Delta x) - f(5)$$
$$= -90 - 43\Delta x - 5(\Delta x)^2 + 90 = -43\Delta x - 5(\Delta x)^2$$
$$= \Delta x\,[-43 - 5\Delta x].$$

Step (iii) $$\frac{\Delta y}{\Delta x} = -43 - 5\Delta x$$

Step (iv) $$m_{\tan} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = -43.$$

10.2.2 Velocity of Rectilinear motion

Suppose an object moves along a straight line according to an equation of motion $s = f(t)$, where s is the displacement (directed distance) of the object from the origin at time t. The function f that describes the motion is called the **position function** of the object. In the time interval from $t = t_0$ to $t = t_0 + \Delta t$, the change in position is $f(t_0 + \Delta t) - f(t_0)$. The average velocity over this time interval is

$$v_{avg} = \frac{\text{displacement}}{\text{time}} = \frac{f(t_0 + \Delta t) - f(t_0)}{\Delta t} = \frac{\text{Change in } s}{\text{Change in } t} = \frac{\Delta s}{\Delta t}$$ which is same as the slope of the secant line PQ in fig. 10.16.

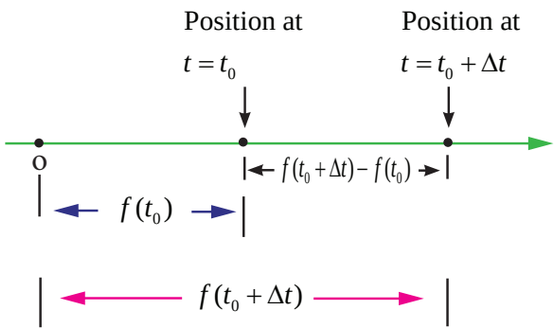


Fig. 10.15

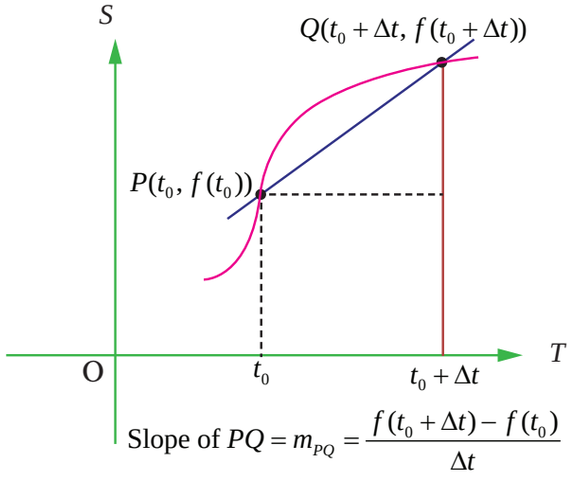


Fig. 10.16

In this time interval Δt (from t_0 to $t_0 + \Delta t$) the motion may be of entirely different types for the same distance covered (traversed). This is illustrated graphically by the fact that we can draw entirely different curves C_1, C_2, C_3 ... between the points P and Q in the plane. These curves are the graphs of quite different motions in the given time intervals, all the motions having the same average velocity $\frac{\Delta s}{\Delta t}$.

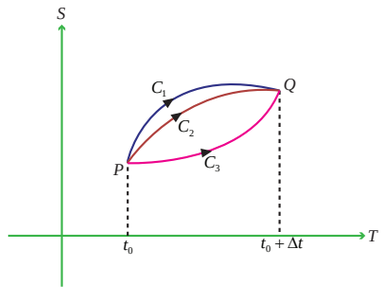


Fig. 10.17

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